

Efficient Transformer-Based Question Answering for Scientific Documents

Abstract

In this paper we propose a novel transformer-based architecture for question answering over scientific documents. Our method, called SciQA-Net, combines a retrieval module with a fine-tuned language model. We evaluate our approach on a custom dataset of 10,000 research paper excerpts and achieve an F1-score of 87.4, outperforming baseline BERT models by 5.2 points.

Introduction

Research papers are dense, technical documents that are time-consuming to read. Automated question answering systems can help researchers quickly extract relevant information. Prior work by Devlin et al. introduced BERT, which revolutionized natural language understanding tasks. However, applying BERT directly to long scientific documents remains challenging due to context length limitations.

Methodology

SciQA-Net consists of two components: a dense retriever built on Sentence-BERT embeddings, and a generator based on a fine-tuned T5 model. Given a question, the retriever first identifies the top-5 most relevant passages using cosine similarity over sentence embeddings. These passages are then concatenated and fed to the generator, which produces a natural language answer grounded in the retrieved context.

Dataset

We constructed a dataset of 10,000 question-answer pairs derived from 500 computer science research papers published between 2018 and 2023. Questions were generated using a combination of automated templates and manual annotation by graduate student volunteers.

Results

Our model achieves an F1-score of 87.4 and an exact-match score of 79.1 on the held-out test set. This represents a 5.2 point improvement over the BERT-large baseline, and a 3.1 point improvement over RoBERTa. We also report a mean answer latency of 340 milliseconds per query, which is fast enough for interactive use.

Limitations

Our approach struggles with questions requiring multi-hop reasoning across distant sections of a paper, and with mathematical notation, which is often poorly represented in the extracted text. Future work should address both of these limitations, potentially through structured document parsing.

Conclusion

We presented SciQA-Net, a retrieval-augmented question answering system for scientific documents, and demonstrated strong performance improvements over existing baselines. We release our dataset and code to encourage further research in this direction.